Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Word page

Enzyme and substrate

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

substrate — The substance that an enzyme acts on.

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

specificity — An enzyme acts on particular substrates. They fit its active site.

denaturation — A change in enzyme shape. This stops the active site binding as it should.

Measure and compare

pH — A measure of how acid or alkaline a substance is.

independent variable — The factor we choose to change.

dependent variable — The outcome we measure.

control variable — A factor we keep the same. This makes the comparison fair.

endpoint — The agreed point we can see and record. We record the time when it is reached.

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Word page · continued

Diagram

Purple shows amylase. Yellow shows starch and smaller sugar products. The shape model shows two moments.

Specificity means an enzyme acts on particular substrates. Amylase breaks starch down into smaller sugars. It speeds this change up. The products leave. The enzyme is not used up. It can act again. This shape model is simple. It does not show real molecules as rigid shapes.

starch substrate

amylase enzyme

smaller sugar products

active site